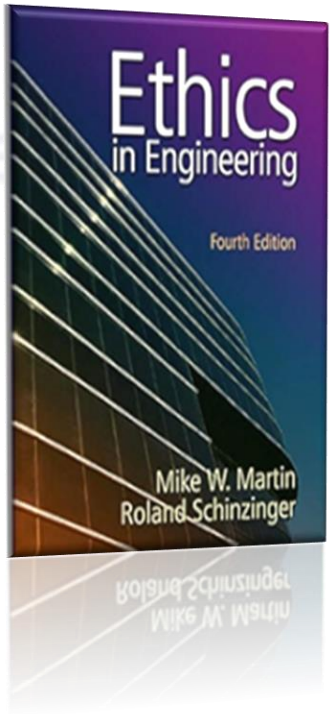
Engineering Ethics (HSS-422) (3+0)

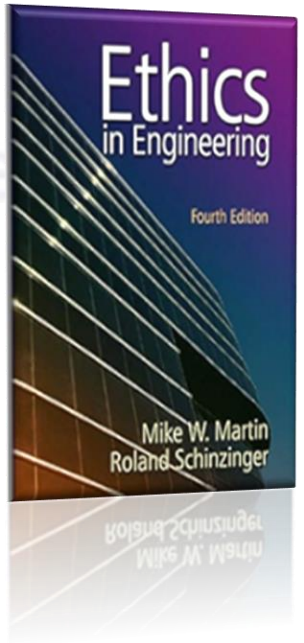






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RISK AND LIABILITY IN ENGINEERING

Week No 09

By Engr. Syed Rizwan Ali

Learning Outcomes

In this Lecture will cover ..

- To Understand Risk Identification Techniques:
- To Assess and Prioritize Risks:
- To Develop Risk Mitigation Strategies:
- To understand Legal and Ethical Responsibilities:
- To Engage with Stakeholders:

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Risk and Liability in Engineering



Understanding of Risk and Liability in Engineering

❖ **Background:**

- Risk and liability in engineering are critical concepts that address the potential for harm, failure, or legal responsibility associated with engineering activities. Here's an introduction to these concepts:
- **Risk in Engineering**
- **Liability in Engineering**
- **Managing Risk and Liability**

Risk in Engineering

Background:

- Risk in engineering refers to the probability and consequence of failure or harm that may result from engineering activities or systems.
- It involves identifying potential hazards, assessing their likelihood and impact, and implementing measures to mitigate these risks.
- Engineering risks can arise from technical failures, environmental factors, human errors, and unforeseen events. Managing these risks is essential to ensure the safety, reliability, and efficiency.

Liability in Engineering

- **Background:**

- Liability in engineering pertains to the legal responsibility that engineers and companies may face if their work fails to meet specified standards or causes harm. This can include personal injury, property damage, financial loss, or environmental impact. Engineers are expected to adhere to high standards of professional conduct and competence. Failure to do so can result in negligence claims, contractual disputes, or regulatory actions.

Managing Risk and Liability

❖ **Background:**

- To manage risk and liability effectively, engineers use various strategies:
- Risk Assessment: Systematically identifying and evaluating risks associated with a project or process.
- Risk Management Plans: Developing strategies to mitigate, transfer, or avoid risks.
- Compliance with Standards: Adhering to industry standards, codes of practice, and legal requirements.

Managing Risk and Liability

❖ **Background:**

- Professional Development: Continuing education to stay updated with technological advancements and regulatory changes.
- Quality Assurance and Control: Implementing systems to ensure that engineering outputs meet required standards.
- Insurance: Obtaining professional liability insurance to cover potential claims.

Real World Examples



Real-World Examples of Risk in Engineering

- **Example: 01**
- **Challenger Space Shuttle Disaster (1986):**
- **Risk:** The failure of an O-ring seal in its right solid rocket booster.
- **Outcome:** The shuttle broke apart 73 seconds into its flight, leading to the death of all seven crew members.
- **Risk Management Issue:** Engineers at NASA had identified potential risks associated with the O-rings in cold weather but failed to delay the launch.

Real-World Examples of Risk in Engineering

- **Example: 02**
- **Deepwater Horizon Oil Spill (2010):**
- **Risk:** A blowout and explosion on the offshore drilling rig, caused by a surge of natural gas.
- **Outcome:** Massive marine and wildlife damage, extensive pollution of the Gulf of Mexico, and significant legal and cleanup costs.
- **Risk Management Issue:** Failures in risk assessment and safety procedures were identified, along with a lack of adequate response strategies for such an event.

Real-World Examples of Liability in Engineering

- **Example: 01**
- **Hyatt Regency Walkway Collapse (1981):**
- **Liability:** Structural collapse of walkways in the Hyatt Regency hotel in Kansas City.
- **Outcome:** Killed 114 people and injured over 200.
- **Legal Consequence:** The engineering firms involved faced severe legal actions due to negligence in design and oversight, resulting in the revocation of licenses for key engineers and significant financial settlements.

Real-World Examples of Liability in Engineering

- **Example: 02**
- **Grenfell Tower Fire (2017):**
- **Liability:** A fire accelerated by flammable cladding material used in a building renovation.
- **Outcome:** 72 people died, and many others were injured.
- **Legal Consequence:** Ongoing investigations and lawsuits have targeted the building management, contractors, and cladding manufacturers, focusing on breaches of building regulations and safety standards.

Real-World Examples of Managing Risk and Liability

- **Example: 01**
- **Fukushima Daiichi Nuclear Disaster (2011):**
- **Risk:** An earthquake and tsunami led to a nuclear meltdown.
- Risk Management Issue: Inadequate safety barriers and preparedness for such a natural disaster.
- **Liability:** TEPCO (Tokyo Electric Power Company) faced intense scrutiny and liability for the disaster management and aftermath, highlighting the importance of disaster preparedness and response in engineering.

Real-World Examples of Managing Risk and Liability

- **Example: 02**
- **Tacoma Narrows Bridge Collapse (1940):**
- **Risk:** Structural failure due to aeroelastic flutter from wind.
- **Outcomes:** The bridge collapsed within months of opening, thankfully without human casualties.
- **Risk Management Issue:** Underestimation of wind effects on the structure during design and construction.

Real-World Examples of Managing Risk and Liability

- **Example: 02**
- **Tacoma Narrows Bridge Collapse (1940):**
- **Learning and Adaptation:** This incident led to changes in engineering practices and standards, particularly in the design and testing of suspension bridges.
- These examples demonstrate how identifying, assessing, and managing risks, and understanding the legal implications of engineering decisions, are crucial for safeguarding lives and properties and for fulfilling professional responsibilities.

The Engineer's Approach To Risk

- **Background:**
- The engineer's approach to risk in the context of "Risk and Liability in Engineering" involves a systematic and disciplined methodology to identify, assess, manage, and mitigate risks throughout the lifecycle of engineering projects. Here's how engineers typically approach risk:
- 1. Risk Identification
- 2. Risk Assessment
- 4. Implementation of Mitigation Strategies
- 5. Monitoring and Review
- 6. Documentation and Reporting
- 7. Legal and Ethical Considerations

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Importance - Engineer's Approach To Risk

- The engineer's approach to risk management is crucial in the field of "Risk and Liability in Engineering" for several key reasons:
- **Enhances Safety and Reduces Incidents:** By proactively identifying and mitigating risks, engineers can significantly reduce the likelihood of accidents and failures. This helps protect public safety, worker health, and environmental integrity, which are paramount in all engineering disciplines.

Importance - Engineer's Approach To Risk

- **Legal Compliance and Reduced Liability:** A systematic approach to risk helps ensure that engineering projects comply with applicable laws, regulations, and standards.
- This not only avoids legal repercussions but also minimizes the potential for costly liabilities due to negligence or failures.

Importance - Engineer's Approach To Risk

- **Improves Project Success and Efficiency:** Effective risk management leads to better project planning and execution.
- By anticipating potential issues and implementing mitigation strategies, projects are less likely to experience delays, cost overruns, and scope changes, leading to more predictable and successful outcomes.

Importance - Engineer's Approach To Risk

- **Maintains Reputation and Trust:** Companies and professionals that effectively manage risks are more likely to be trusted by clients, investors, and the public. This trust is crucial for maintaining a positive reputation and securing future business opportunities.
- **Facilitates Innovation and Competitiveness:** Understanding and managing risks allows engineers to explore innovative solutions and technologies with confidence. By balancing

Importance - Engineer's Approach To Risk

- **Facilitates Innovation and Competitiveness:** creativity with risk assessment, engineers can push the boundaries of what is possible while still ensuring project viability and safety.

The Public's Approach to Risk

- **Background:**
- The public's approach to risk in the context of "Risk and Liability in Engineering" is distinct from that of engineers, as it focuses more on perception, trust, and societal impact. Here's how the general public typically approaches and perceives risk in engineering projects:
- 1. Risk Perception
- 2. Trust in Institutions and Professionals
- 3. Communication and Information Access
- 4. Impact on Community and Environment
- 5. Advocacy and Activism

Case Study

❖ Case Study: Government Regulator's Approach to Risk in Engineering - The Aliso Canyon Gas Leak:

- In October 2015, the Aliso Canyon natural gas storage facility in California experienced one of the largest methane leaks in U.S. history. The leak not only posed significant environmental and health risks but also highlighted the need for stringent regulatory oversight in engineering projects.

Case Study

- **Government Response and Risk Management:**
- The government regulator, the California Public Utilities Commission (CPUC), along with the California Department of Conservation's Division of Oil, Gas, and Geothermal Resources (DOGGR), responded by conducting a thorough investigation into the causes of the leak.
- Their approach to managing this risk involved several key steps:
- 1. Immediate Risk Mitigation:
- 2. Technical Analysis and Fault Identification:
- 3. Regulatory Overhaul:

Case Study

- **Government Response and Risk Management:**
- Their approach to managing this risk involved several key steps:
- 1. Immediate Risk Mitigation:
- 2. Technical Analysis and Fault Identification:
- 3. Regulatory Overhaul:
- 4. Long-term Monitoring and Compliance:
- 5. Public Engagement and Transparency:
- This case illustrates the government regulator's role in managing risk through immediate action, regulatory improvement, and public engagement, ensuring that such risks are minimized and that similar incidents are prevented in the future.

Case Study

- **Government Response and Risk Management:**
- **Immediate Risk Mitigation:** The regulators coordinated with Southern California Gas Company (SoCalGas) to stop the leak and mitigate its immediate effects, ensuring public safety and minimizing environmental impact.
- **Technical Analysis and Fault Identification:** An extensive technical analysis was undertaken to understand the failure mechanisms. It was determined that the well's casing failure was due

Case Study

- **Technical Analysis and Fault Identification:**
- It was determined that the well's casing failure was due to microbial corrosion, and the safety valves were not adequately maintained, having been removed in 1979 without replacement.
- **Regulatory Overhaul:** In response to the findings, the CPUC and DOGGR revised the regulatory framework governing gas storage facilities. New safety standards were implemented, including mandatory inspections, risk assessments at all gas storage facilities, and enhanced emergency response protocols.

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Case Study

- **Government Response and Risk Management:**
- **Long-term Monitoring and Compliance:** The regulators established a long-term monitoring system and set strict compliance requirements for SoCalGas and other similar facilities.
- These measures were designed to prevent future incidents by ensuring ongoing oversight and accountability.

Case Study

- **Government Response and Risk Management:**
- **Public Engagement and Transparency:** Recognizing the public's concern and the need for transparency, the CPUC held multiple public hearings and provided regular updates on their findings and the steps being taken to prevent future leaks.
- This case illustrates the government regulator's role in managing risk through immediate action, regulatory improvement, and public engagement, ensuring that such risks are minimized and that similar incidents are prevented in the future.

Thanks

Any Question

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